

Literature Answer: The Hyperbolic-Space Free Space Question of Doucha–Kaufmann

Run `fa_banach_001`, agent `agent_lane_00`

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Status

Packet type: literature answer already present in later literature.

Verdict: Gartland’s arXiv:2406.10986 proves

$$\mathrm{LF}(\mathbb{H}^n) \simeq \mathrm{LF}(\mathbb{R}^n)$$

for hyperbolic n -space. This gives a positive answer to the isomorphism clause of Doucha–Kaufmann’s Question 6. Since $\mathcal{F}(\mathbb{R}^n)$ has a Schauder basis by Hájek–Pernecká, it also gives a positive answer to the Schauder-basis clause.

Source Metadata

Original source. Michal Doucha and Pedro L. Kaufmann, *Approximation properties in Lipschitz-free spaces over groups*, arXiv:2005.09785v2.

Supporting answer source. Chris Gartland, *Hyperbolic Metric Spaces and Stochastic Embeddings*, arXiv:2406.10986v2.

Basis input. Petr Hájek and Eva Pernecká, *On Schauder bases in Lipschitz-free spaces*, J. Math. Anal. Appl. 416 (2014), 629–646.

Open Question

After proving that every net $N \subseteq \mathbb{H}^n$ has $\mathcal{F}(N)$ with a Schauder basis, Doucha–Kaufmann ask:

Does $\mathcal{F}(\mathbb{H}^n)$ have the Schauder basis? Do we have $\mathcal{F}(\mathbb{H}^n) \simeq \mathcal{F}(\mathbb{R}^n)$?

In Corollary 4.4, we have shown that for any net $N \subseteq \mathbb{H}^n$, $\mathcal{F}(N)$ has a Schauder basis. Given the prominence of the space $\mathbb{H}^n$ in geometry and beyond, it is important to understand the Lipschitz-free space of $\mathbb{H}^n$ itself. We also note that the hyperbolic spaces $\mathbb{H}^n$ together with the Euclidean spaces $\mathbb{R}^n$ and Euclidean spheres $\mathbb{S}^n$ are important as the model spaces of spaces of constant curvature (see e.g. [4] for a thorough treatment). Since for $\mathbb{R}^n$ and $\mathbb{S}^n$ with their canonical Euclidean metrics we have by [1, Theorem 4.21], $\mathcal{F}(\mathbb{R}^n) \simeq \mathcal{F}(\mathbb{S}^n)$, the answer to the following would be desirable.

Question 6. *Does $\mathcal{F}(\mathbb{H}^n)$ have the Schauder basis? Do we have $\mathcal{F}(\mathbb{H}^n) \simeq \mathcal{F}(\mathbb{R}^n)$?*

We note that the previous question is related to the stronger version of Question 3 that considers a general Riemannian manifold since $\mathbb{H}^n$ is a Riemannian manifold with

Supporting Answer

Gartland's introduction states the answer as Theorem D, citing Corollary 6.4:

embeds into a pointed $\mathbb{R}$ -tree and $\mathrm{LF}(A) \approx \ell^+$.

As an application of Theorem C, we show through Corollary 6.4 in §6 that the free space over hyperbolic n -space $\mathbb{H}^n$ is isomorphic to the free space over Euclidean n -space $\mathbb{R}^n$. See §6 for background on the space $\mathbb{H}^n$.

Theorem D (Corollary 6.4). $\mathrm{LF}(\mathbb{H}^n) \approx \mathrm{LF}(\mathbb{R}^n)$.

The proof of Theorem D can be summarized as follows: we establish in Lemma 6.2 a general local-to-global criterion for finite Nagata-dimensional spaces X , which states that $\mathrm{LF}(X)$ is isomorphic to a complemented subspace of the ℓ^1 -sum $(\oplus_{a \in A}^1 \mathrm{LF}(B_r(a))) \oplus \mathrm{LF}(A)$, where $0 < r < \infty$ is some fixed radius and $A \subseteq X$ is a uniformly discrete subset. We can then apply this lemma to $X = \mathbb{H}^n$ and use the fact that any infinite, uniformly discrete $A \subset \mathbb{H}^n$ satisfies the hypotheses of Theorem C to obtain that $\mathrm{LF}(\mathbb{H}^n)$

The formal statement and proof appear as Corollary 6.4.

Combining the preceding results yields the main result of this section, which proves Theorem D.

Corollary 6.4. $\mathrm{LF}(\mathbb{H}^n) \approx \mathrm{LF}(\mathbb{R}^n)$.

Proof. We will show that the hypotheses of all three items of Theorem 6.3 are satisfied with $X = \mathbb{H}^n$ and $G = \mathbb{R}^n$. Indeed, the hypotheses of items (1) and (2) are satisfied by virtue of the fact that $\mathbb{H}^n$ is a homogeneous n -dimensional Riemannian manifold. It remains to show that $\mathrm{LF}(A) \xhookrightarrow{c} \mathrm{LF}(\mathbb{R}^n)$ for every uniformly discrete subset $A \subseteq \mathbb{H}^n$.

Let $A \subseteq \mathbb{H}^n$ be uniformly discrete. Then A is locally finite since $\mathbb{H}^n$ is proper. If A is

Formal Statement

Theorem. For every integer $n \geq 2$,

$$\mathcal{F}(\mathbb{H}^n) \simeq \mathcal{F}(\mathbb{R}^n).$$

Consequently, $\mathcal{F}(\mathbb{H}^n)$ has a Schauder basis.

Proof. Gartland proves in [2, Corollary 6.4] that $\mathrm{LF}(\mathbb{H}^n) \simeq \mathrm{LF}(\mathbb{R}^n)$. Gartland's notation $\mathrm{LF}(M)$ denotes the Lipschitz-free space over M , the same object denoted $\mathcal{F}(M)$ by Doucha–Kaufmann. Thus $\mathcal{F}(\mathbb{H}^n) \simeq \mathcal{F}(\mathbb{R}^n)$.

By Hájek–Pernecká [3], $\mathcal{F}(\mathbb{R}^n)$ has a Schauder basis. Since the existence of a Schauder basis is preserved under Banach-space isomorphism, $\mathcal{F}(\mathbb{H}^n)$ has a Schauder basis as well. $\square$

Scope And Limitations

Gartland states the result for $2 \leq n \in \mathbb{N}$, the standard range for real hyperbolic n -space in the cited proof. The $n = 1$ edge case is trivial because the hyperbolic line is isometric to $\mathbb{R}$.

The same source paper also reiterates the Candido–Cuth–Doucha question about whether finitely generated hyperbolic groups have free space isomorphic to ℓ_1 . That adjacent question is already recorded in this run as `solutions/literature_already_answered/1809.09957_hyperbolic_groups_free_space_l1/` and is not duplicated here.

Search And Verification Notes

The run indexes were searched for arXiv:2005.09785, arXiv:2406.10986, hyperbolic-space phrases, and the core keywords “Schauder basis”, “Lipschitz-free”, and “hyperbolic group”. No previous packet for the $\mathbb{H}^n$ clause was found. A separate existing packet already covers the hyperbolic-group ℓ_1 clause.

References

- [1] M. Doucha and P. L. Kaufmann, *Approximation properties in Lipschitz-free spaces over groups*, arXiv:2005.09785.
- [2] C. Gartland, *Hyperbolic Metric Spaces and Stochastic Embeddings*, arXiv:2406.10986.
- [3] P. Hájek and E. Pernecká, *On Schauder bases in Lipschitz-free spaces*, J. Math. Anal. Appl. 416 (2014), 629–646.